calc_num_exercicios_06_newton_raphson

October 25, 2026

1 Lista de Exercícios 06 – Raízes de Equações – Newton-Raphson

Aluna: Maria Eduarda Saldanha Alves - 25/10/2026

1.1 1 - Utilize o método de Newton-Raphson para encontre a raiz aproximada das seguintes equações. Considere os seguintes pontos:

- Precisão de 6 casas decimais;
- Número máximo de 50 iterações e $\epsilon < 10^{-5}$;
- Isole o intervalo da raiz;
- Analise a convergência do método;
- Apresente o código, solução e passo a passo para chegar nas respostas.

[275]:
```
import numpy as np
import sympy as sy
import matplotlib.pyplot as pyplot
from IPython.display import Math
from newton_raphson import newton_raphson

def plot(fx, intervalo):
    xx = np.linspace(intervalo[0], intervalo[1])
    pyplot.plot(xx, fx(xx), label='f(x)')
    pyplot.grid()
    pyplot.show()

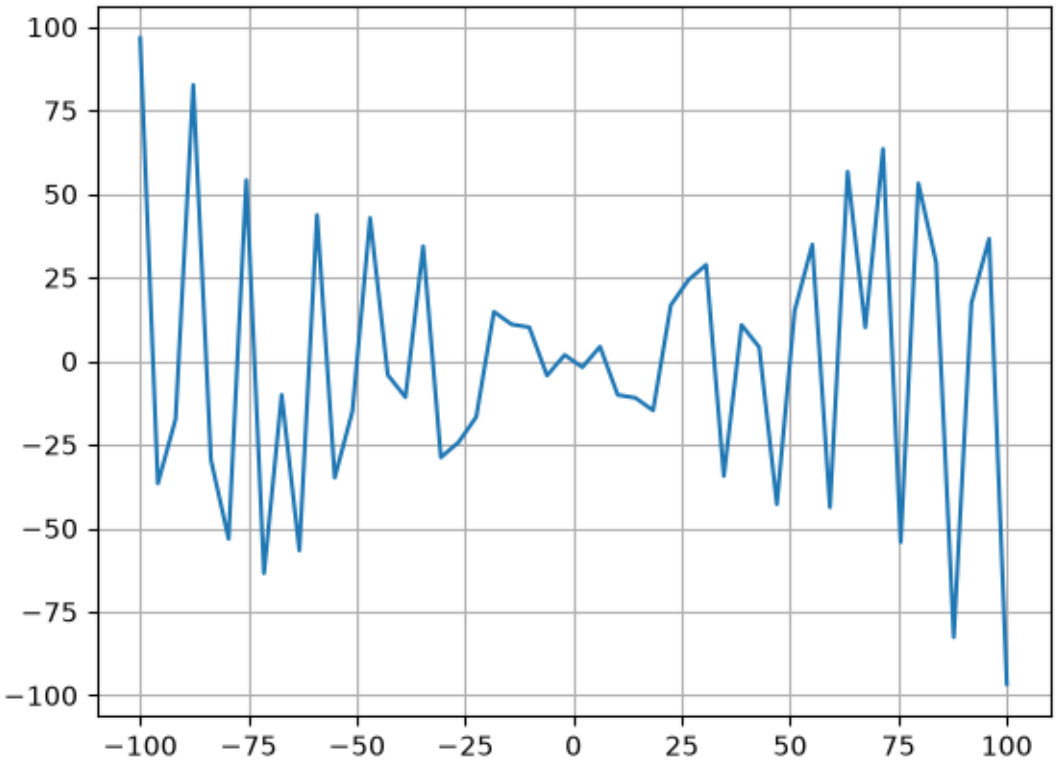
def plot3(fx_, intervalo):
    x_ = sy.Symbol('x')
    flx_ = sy.diff(fx_, x_)
    fx = sy.lambdify(x_, fx_)
    flx = sy.lambdify(x_, flx_)
    f2lx_ = sy.diff(flx_, x_)
    f2lx = sy.lambdify(x_, f2lx_)
    xx = np.linspace(intervalo[0], intervalo[1])
    pyplot.plot(xx, fx(xx), label='f(x)')
    pyplot.plot(xx, flx(xx), label='f\'(x)')
    pyplot.plot(xx, f2lx(xx), label='f\'\'(x)')
    pyplot.legend()
    pyplot.grid()
```

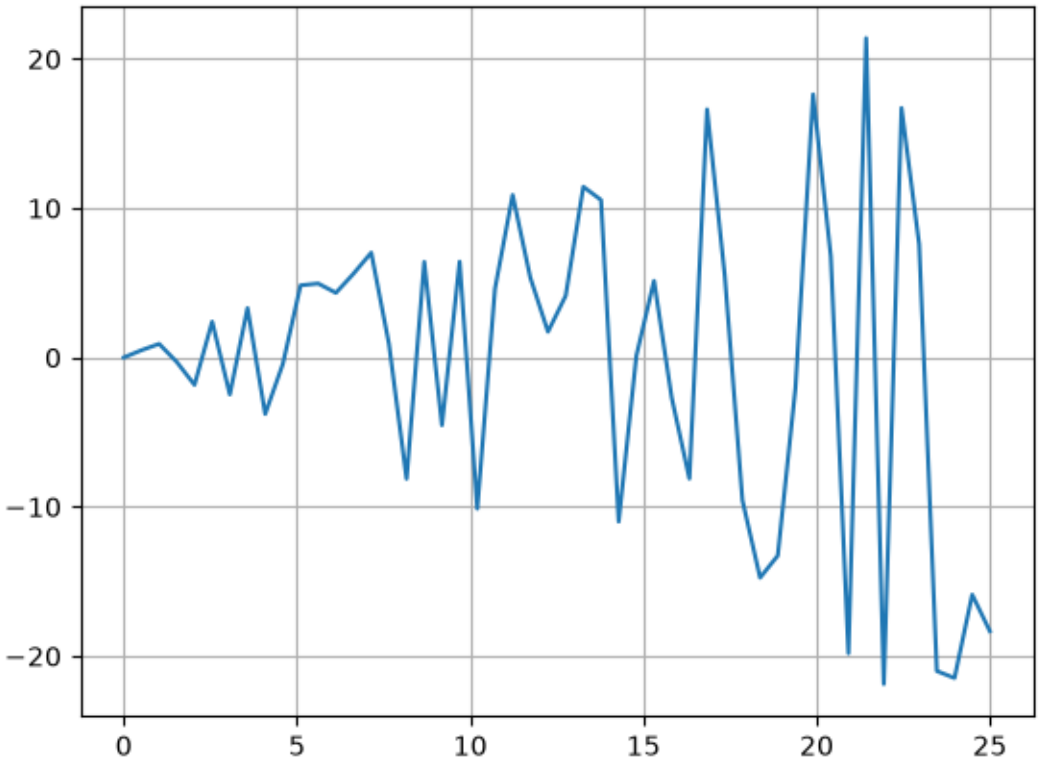
```
    pyplot.show()
```

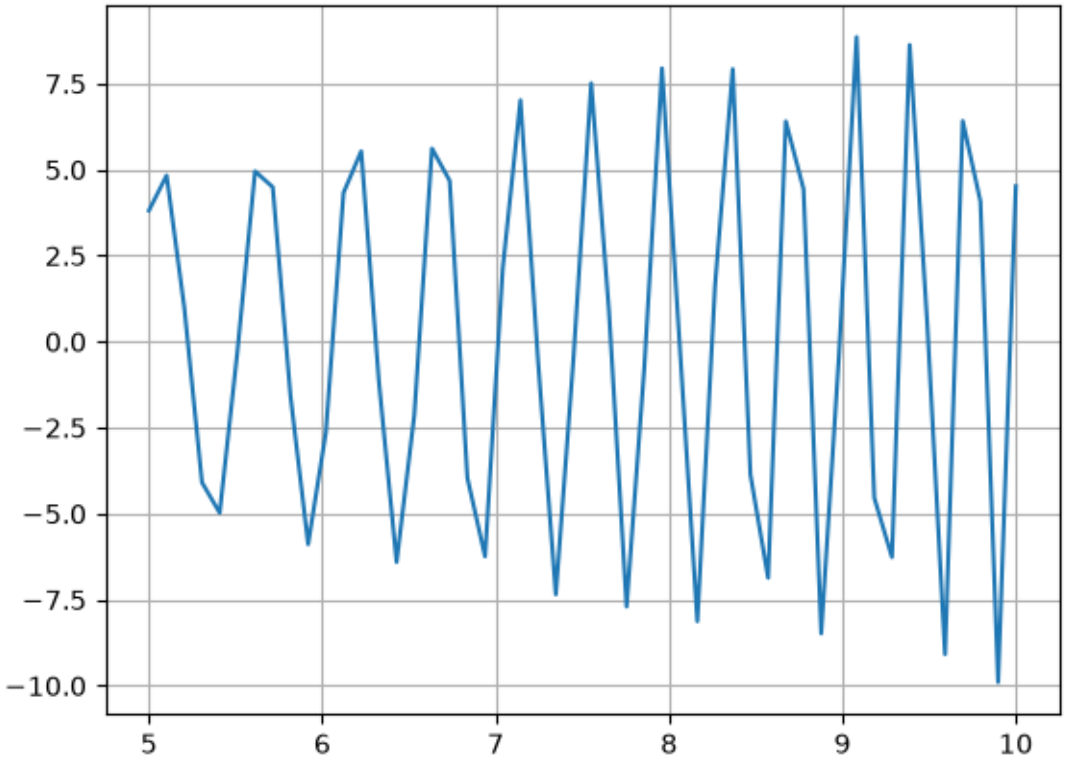
1.1.1 a) $f(x) = x \sin(x^2 + 1)$

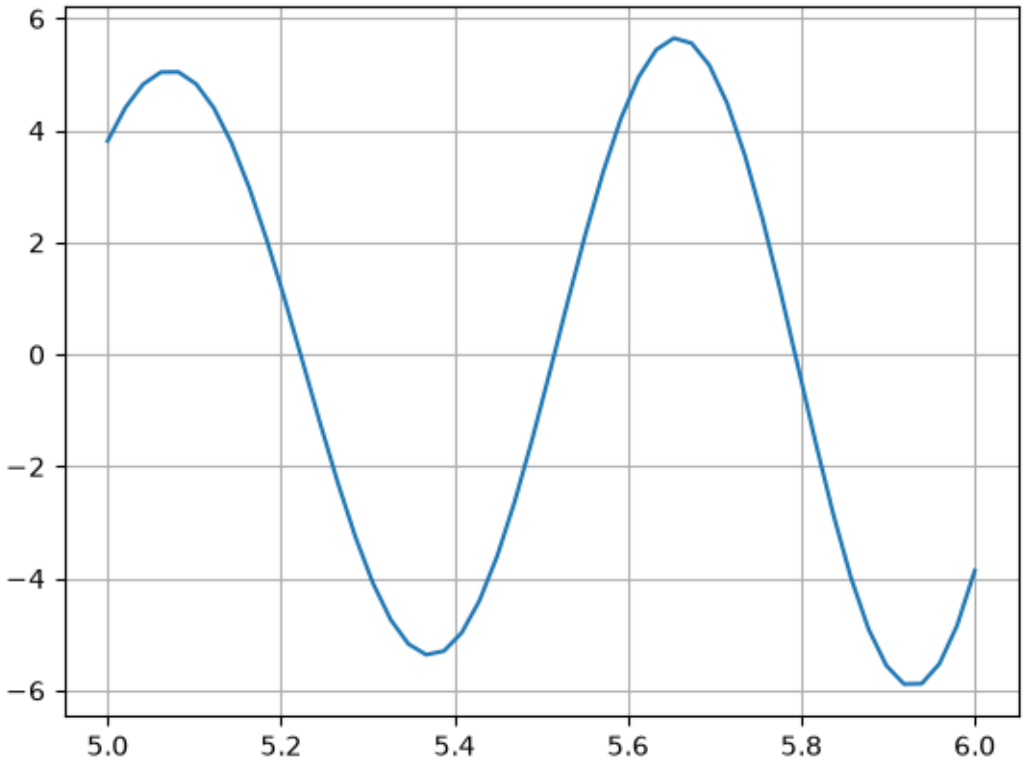
[276]:
```
# Definição do intervalo
x = sy.Symbol('x')
fx_ = x * sy.sin(x**2+1)
display(Math('f(x) = ' + sy.latex(fx_)))
fx = sy.lambdify(x, fx_)
plot(fx, [-100, 100])
plot(fx, [0, 25])
plot(fx, [5, 10])
plot(fx, [5, 6])
plot(fx, [5, 5.5])
```

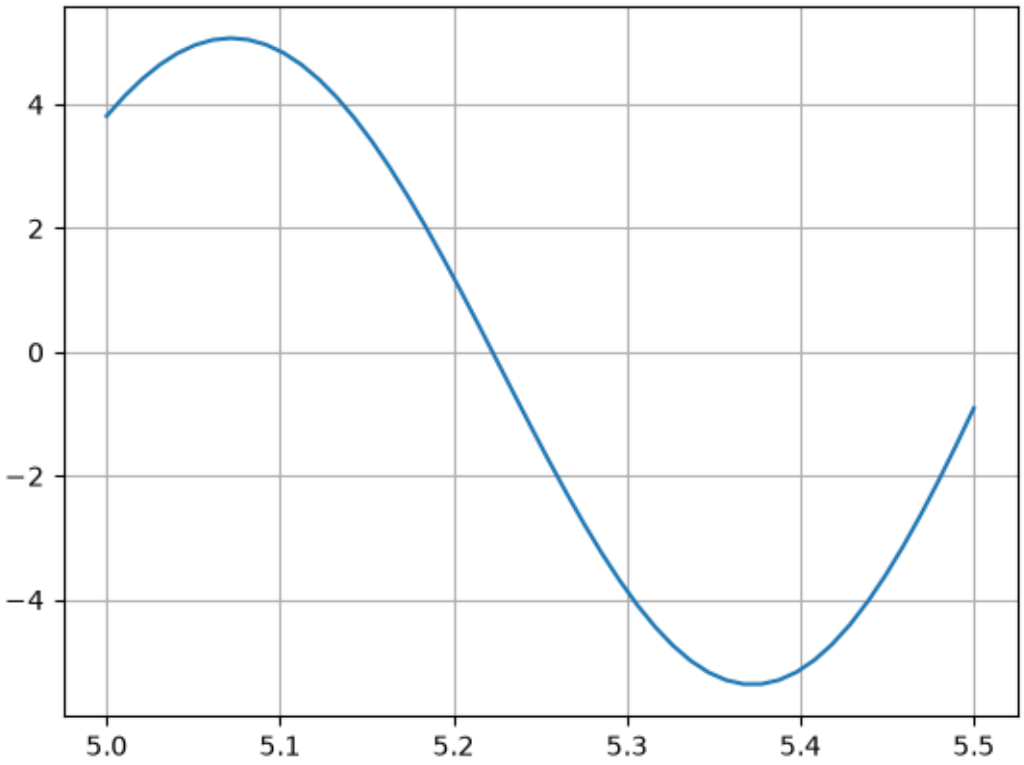
$f(x) = x \sin(x^2 + 1)$



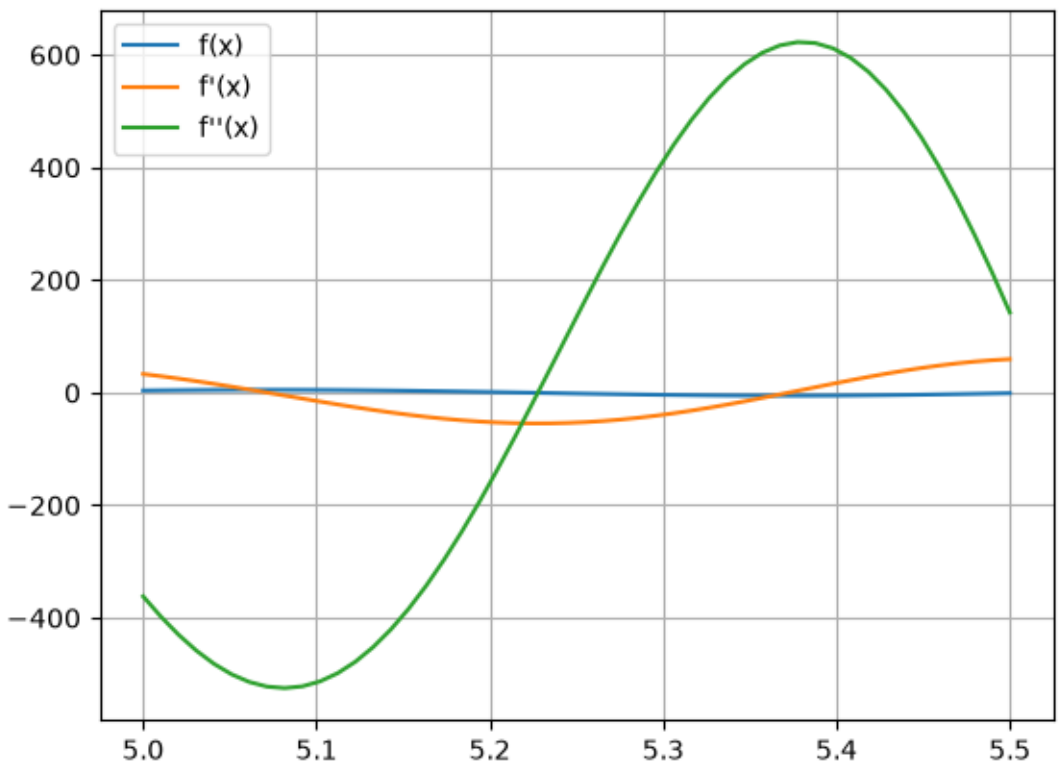

20
10
0
−10
−20
0
5
10
15
20
25


7.5
5.0
2.5
0.0
−2.5
−5.0
−7.5
−10.0
5
6
7
8
9
10


6
4
2
0
−2
−4
−6
5.0
5.2
5.4
5.6
5.8
6.0



[277]:
```
intervalo = [5, 5.5]
plot3(fx_, intervalo)
```



[278]:
```
# Teste de convergência

# 1) f(x) é contínua com raiz no intervalo [a, b]
# 2) f'(x) \= 0 para todo x em [a, b]
# 3) f''(x) não muda de sinal em [a, b]

# Não passa pelo teste de convergência, mas vamos executar o método
raiz, iteracoes = newton_raphson(fx_, intervalo, 10e-5, 50)
display(iteracoes)
```

	k	x	f(x)	epsilon
0	1	5.250000	-1.492021e+00	NaN
1	2	5.221921	3.068785e-02	5.377173e-03
2	3	5.222484	-5.314994e-06	1.077589e-04
3	4	5.222483	-1.797838e-13	1.865699e-08

[279]:
```
print('Raiz:', raiz)
# O método convergiu para a raiz
```

```
Raiz: 5.222483497562072
```

1.1.2 b) $f(x) = e^{\sin(x)} - 1$

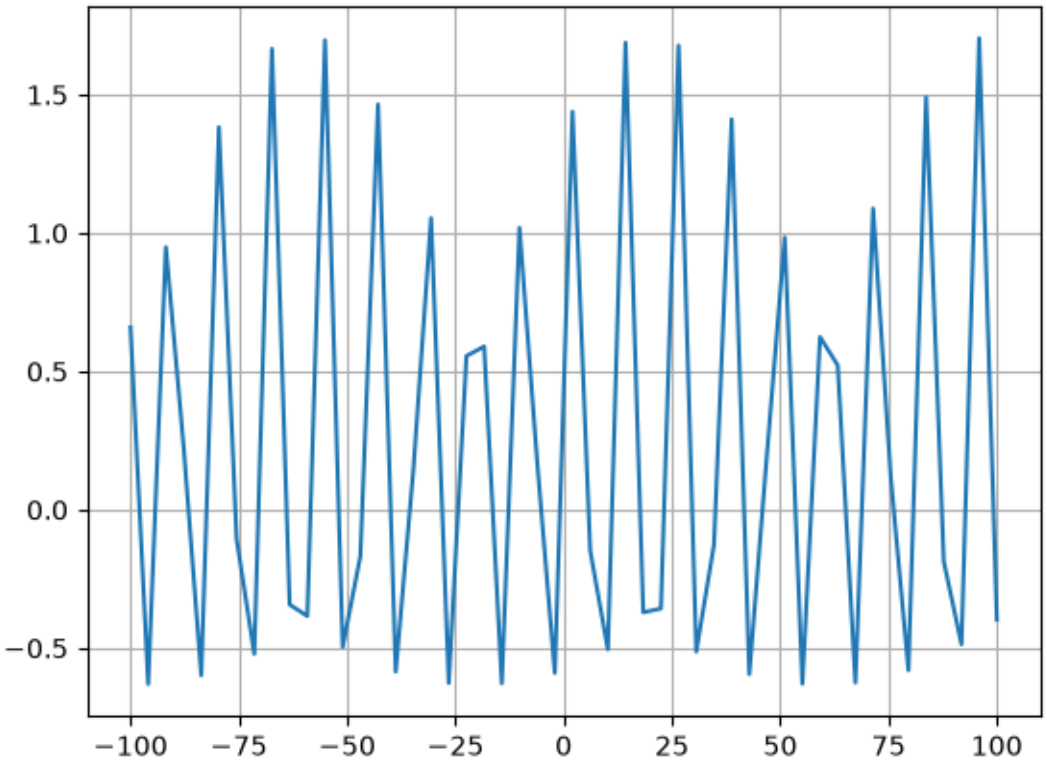
[280]:
```
# Definição do intervalo
x = sy.Symbol('x')
fx_ = sy.exp(sy.sin(x)) - 1

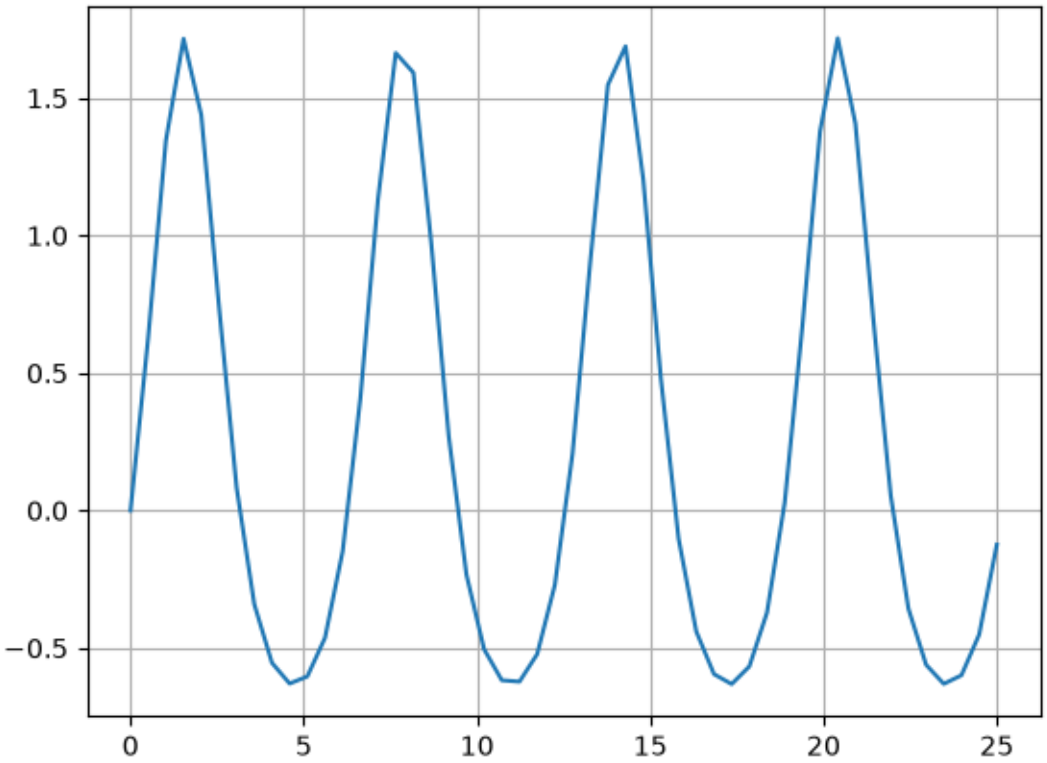
display(Math('f(x) = ' + sy.latex(fx_)))

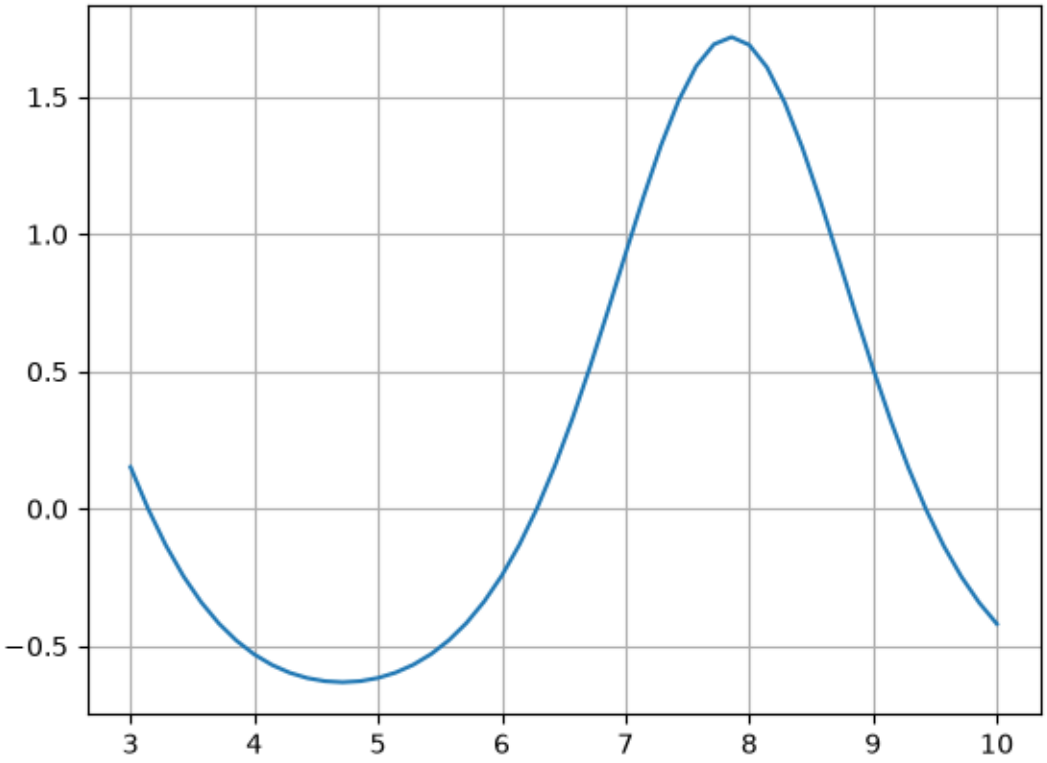
fx = sy.lambdify(x, fx_)

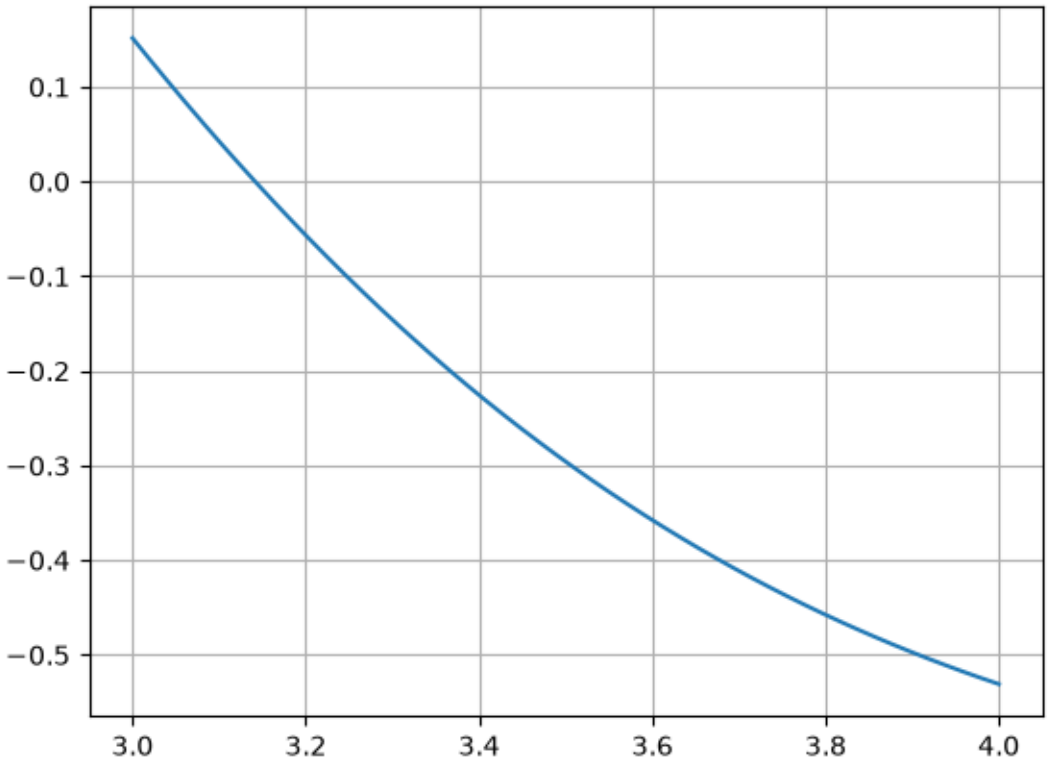
plot(fx, [-100, 100])
plot(fx, [0, 25])
plot(fx, [3, 10])
plot(fx, [3, 4])
plot(fx, [3, 3.5])
```

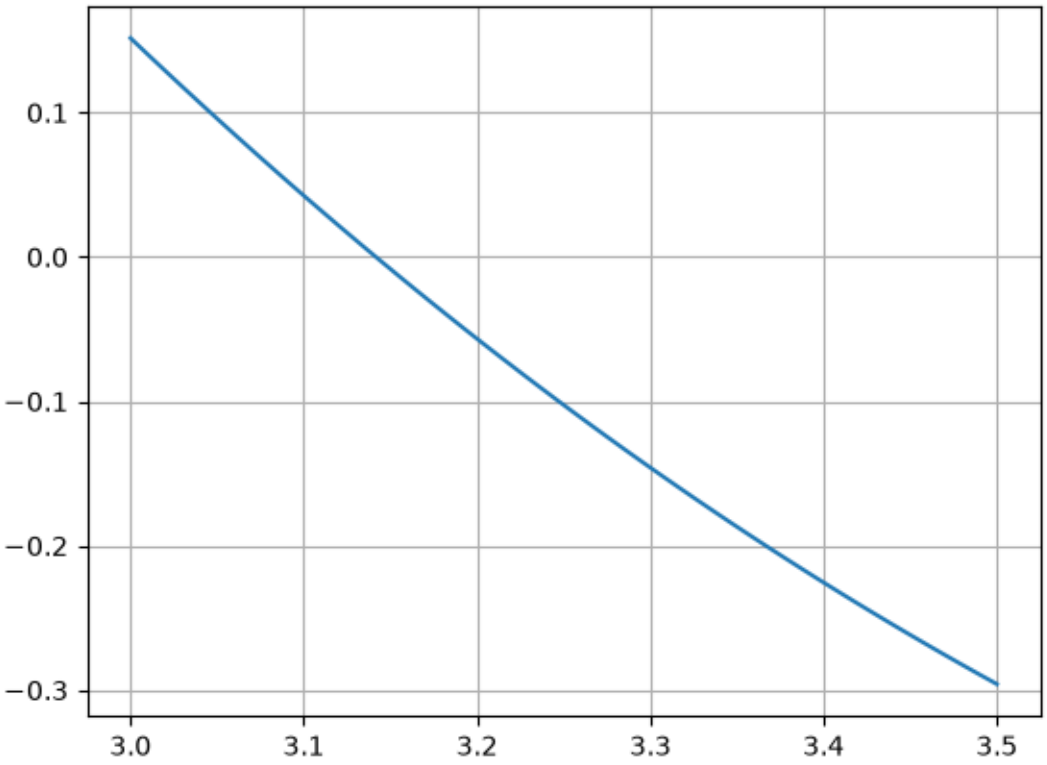
$f(x) = e^{\sin(x)} - 1$



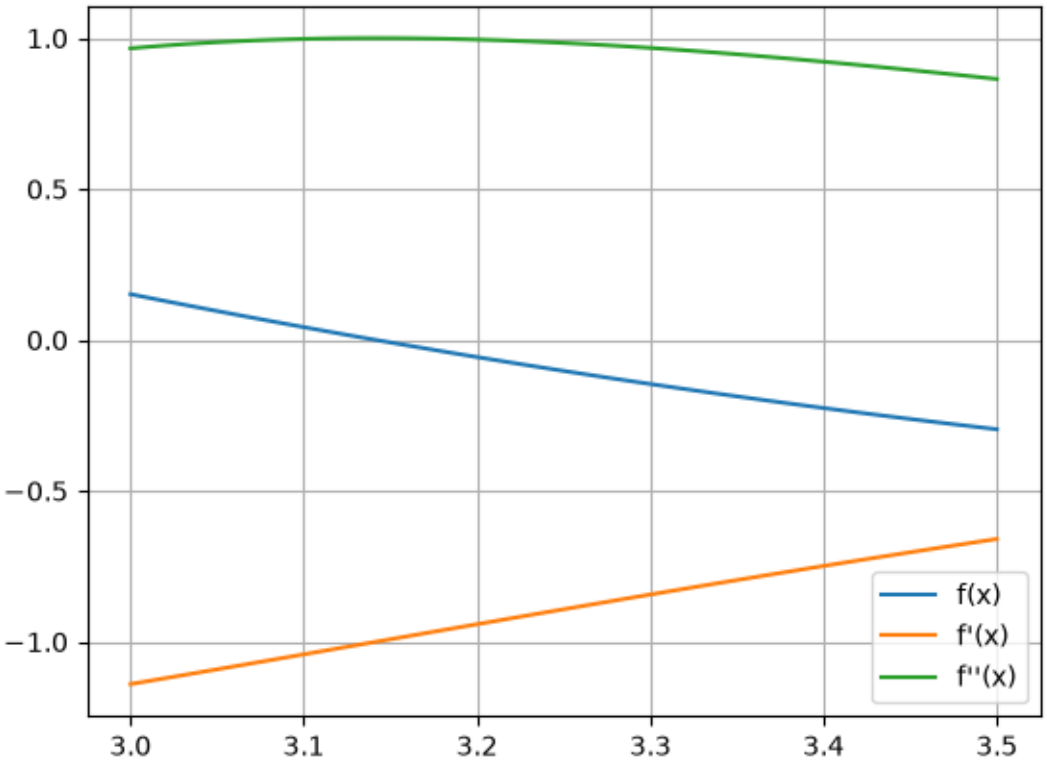

1.5
1.0
0.5
0.0
−0.5
0
5
10
15
20
25


1.5
1.0
0.5
0.0
−0.5
3
4
5
6
7
8
9
10


0.1
0.0
−0.1
−0.2
−0.3
−0.4
−0.5
3.0
3.2
3.4
3.6
3.8
4.0



[281]:
```
intervalo = [3, 3.5]
plot3(fx_, intervalo)
```



[282]:
```
# Teste de convergência

# 1) f(x) é contínua com raiz no intervalo [a, b]
# 2) f'(x) != 0 para todo x em [a, b]
# 3) f''(x) não muda de sinal em [a, b]

# Execução do método
raiz, iteracoes = newton_raphson(fx_, intervalo, 1e-5, 50)

display(iteracoes)

print('Raiz:', raiz)
# O método convergiu para a raiz
```

	k	x	f(x)	epsilon
0	1	3.250000	-1.025475e-01	NaN
1	2	3.135060	6.553881e-03	0.036663
2	3	3.141571	2.119813e-05	0.002073
3	4	3.141593	2.246712e-10	0.000007

```
Raiz: 3.141592653365122
```

1.1.3 c) $f(x) = 2x^4 - 15x^3 + x - 1$

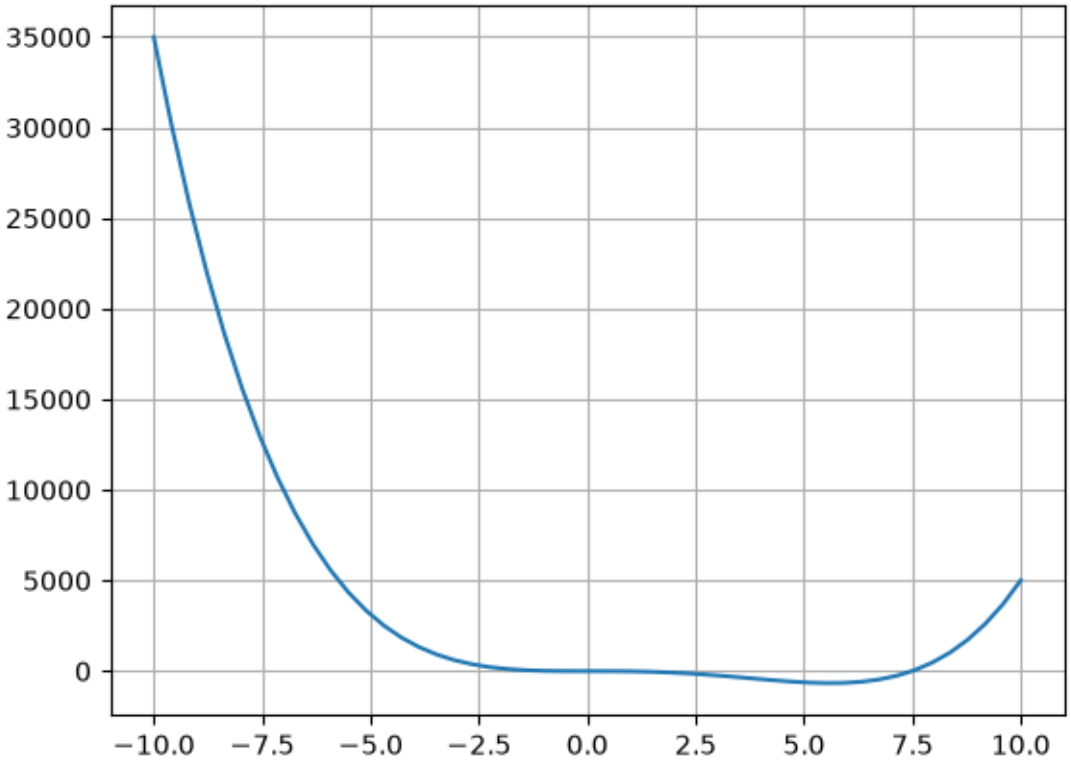
[283]:
```
# Definição da função
x = sy.Symbol('x')
fx_ = 2*x**4 - 15*x**3 + x - 1

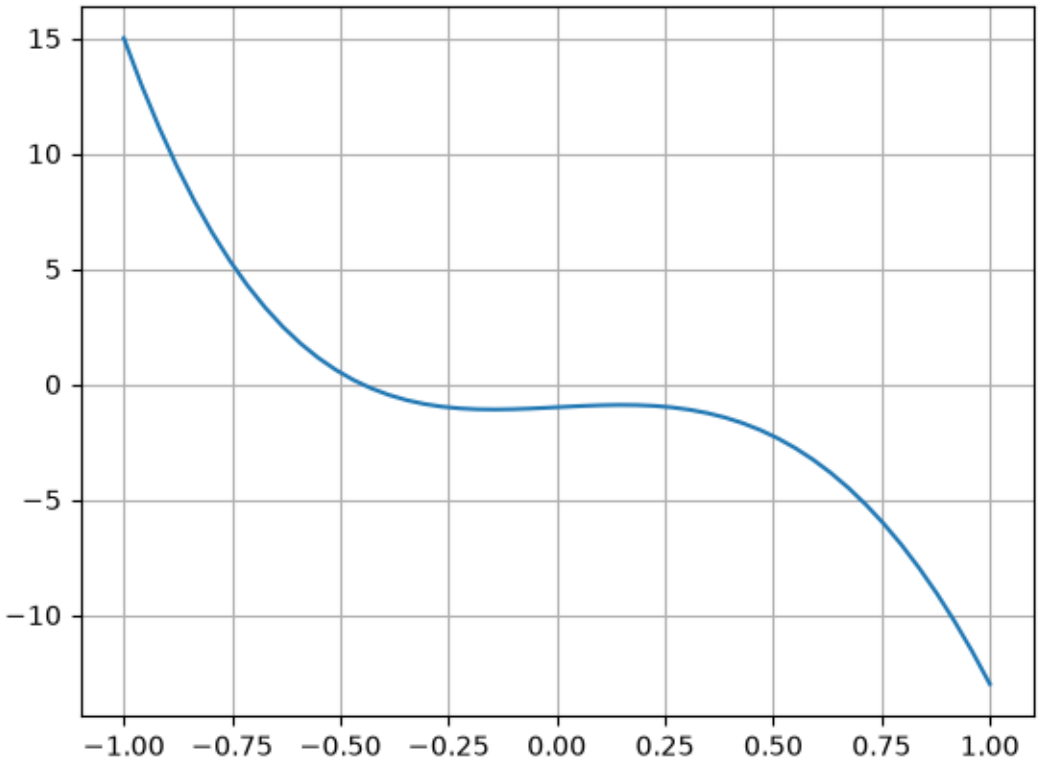
display(Math('f(x) = ' + sy.latex(fx_)))

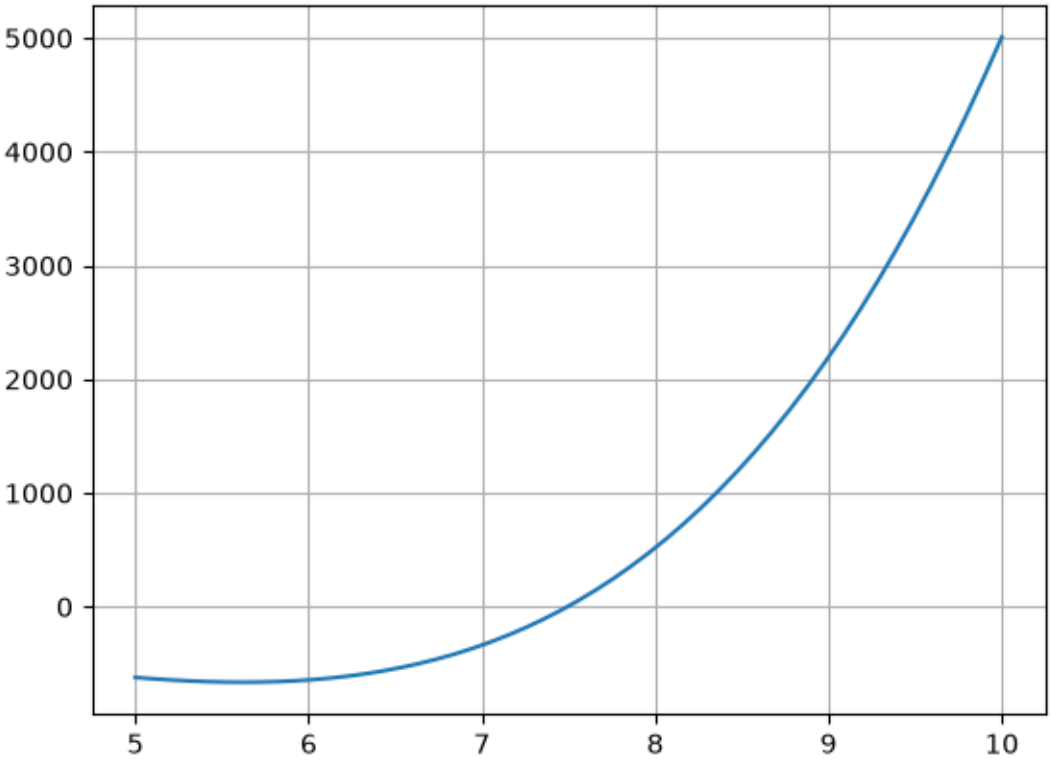
fx = sy.lambdify(x, fx_)

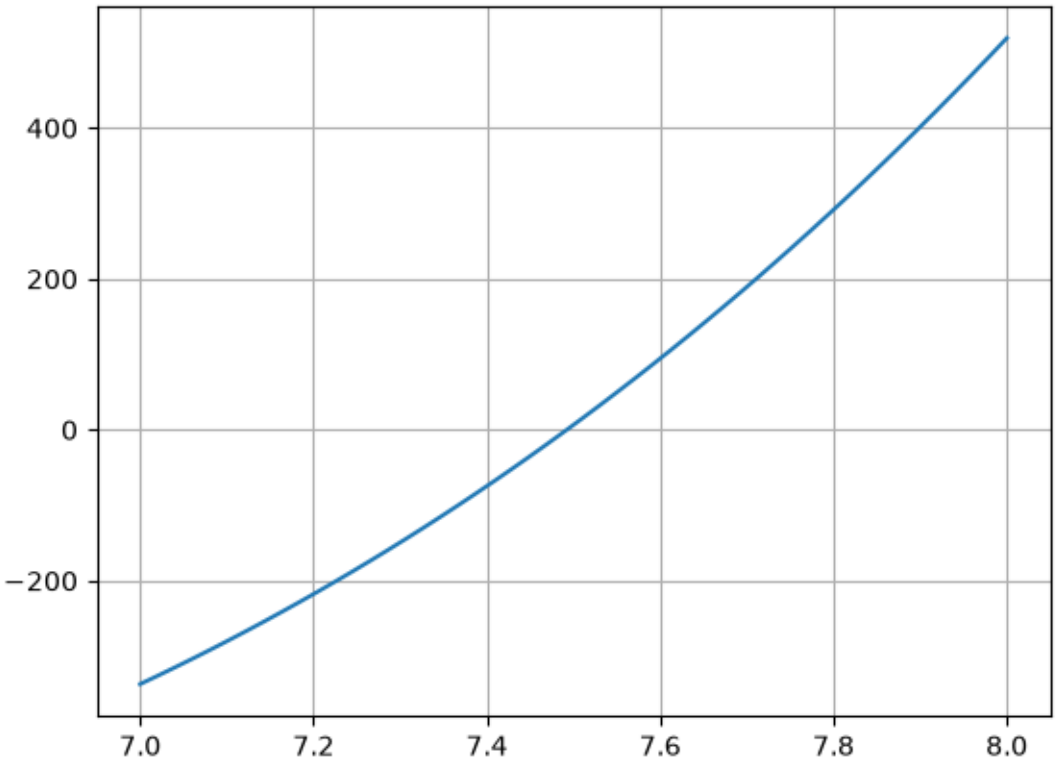
plot(fx, [-10, 10])
plot(fx, [-1, 1])
plot(fx, [5, 10])
plot(fx, [7, 8])
```

$f(x) = 2x^4 - 15x^3 + x - 1$

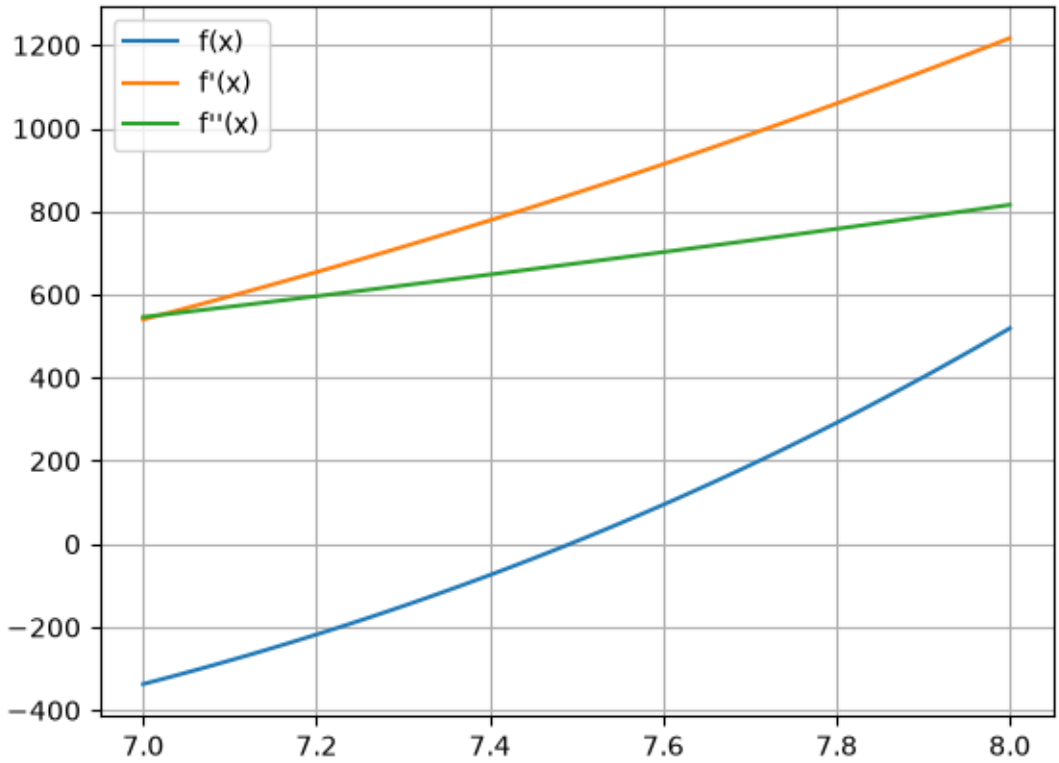



15
10
5
0
−5
−10
−1.00
−0.75
−0.50
−0.25
0.00
0.25
0.50
0.75
1.00


5000
4000
3000
2000
1000
0
5
6
7
8
9
10



[284]:
```
intervalo = [7, 8]
plot3(fx_, intervalo)
```



[285]:
```
# Teste de convergência

# 1) f(x) é contínua com raiz no intervalo [a, b]
# 2) f'(x) != 0 para todo x em [a, b]
# 3) f''(x) não muda de sinal em [a, b]

# Passa pelo teste de convergência
raiz, iteracoes = newton_raphson(fx_, intervalo, 1e-5, 50)

display(iteracoes)

print('Raiz:', raiz)
# O método convergiu para a raiz
```

	k	x	f(x)	epsilon
0	1	7.500000	6.500000e+00	NaN
1	2	7.492305	1.996174e-02	0.001027
2	3	7.492282	1.902052e-07	0.000003

```
Raiz: 7.492281639487262
```

1.1.4 d) $f(x) = \dfrac{(2x^3 - 10x)\cos(x)}{x^2 - 5}$

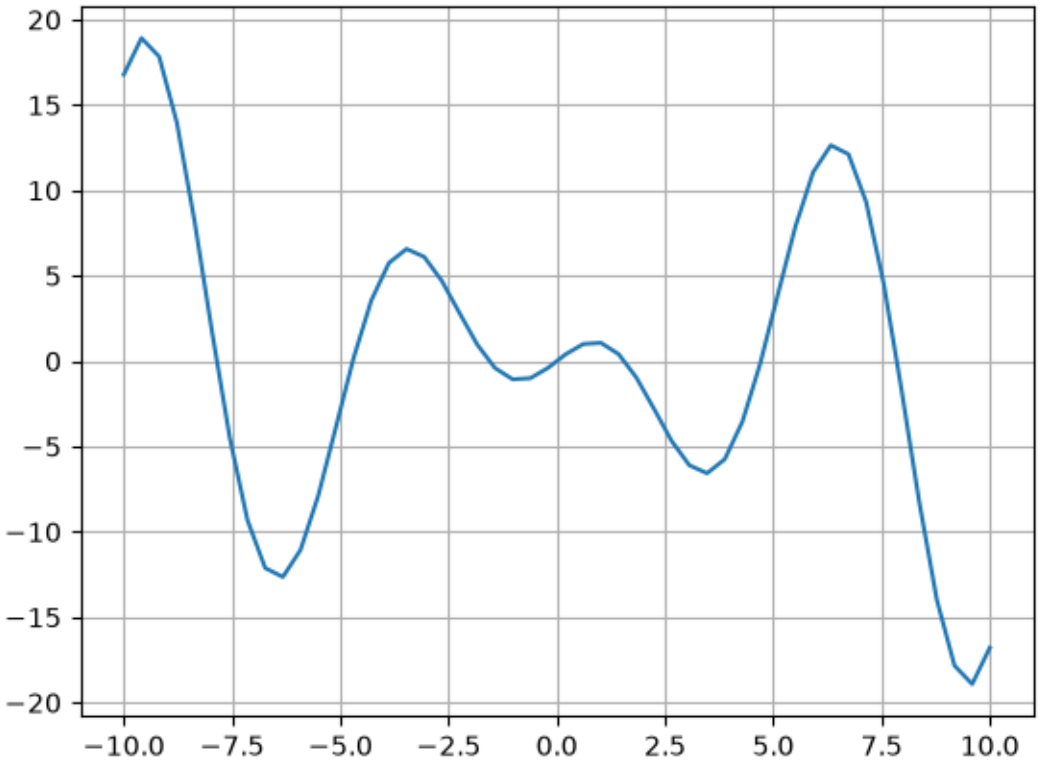
[286]:
```
# Definição da função
x = sy.Symbol('x')
fx_ = ((2*x**3 - 10*x)*sy.cos(x))/(x**2 - 5)

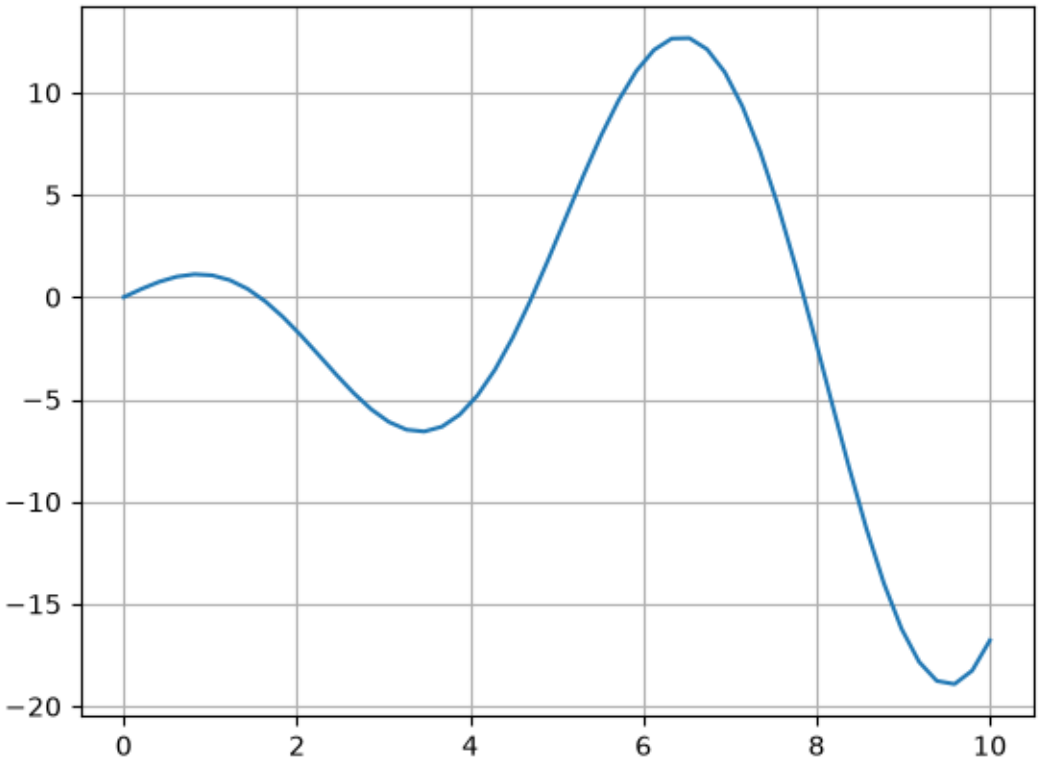
display(Math('f(x) = ' + sy.latex(fx_)))

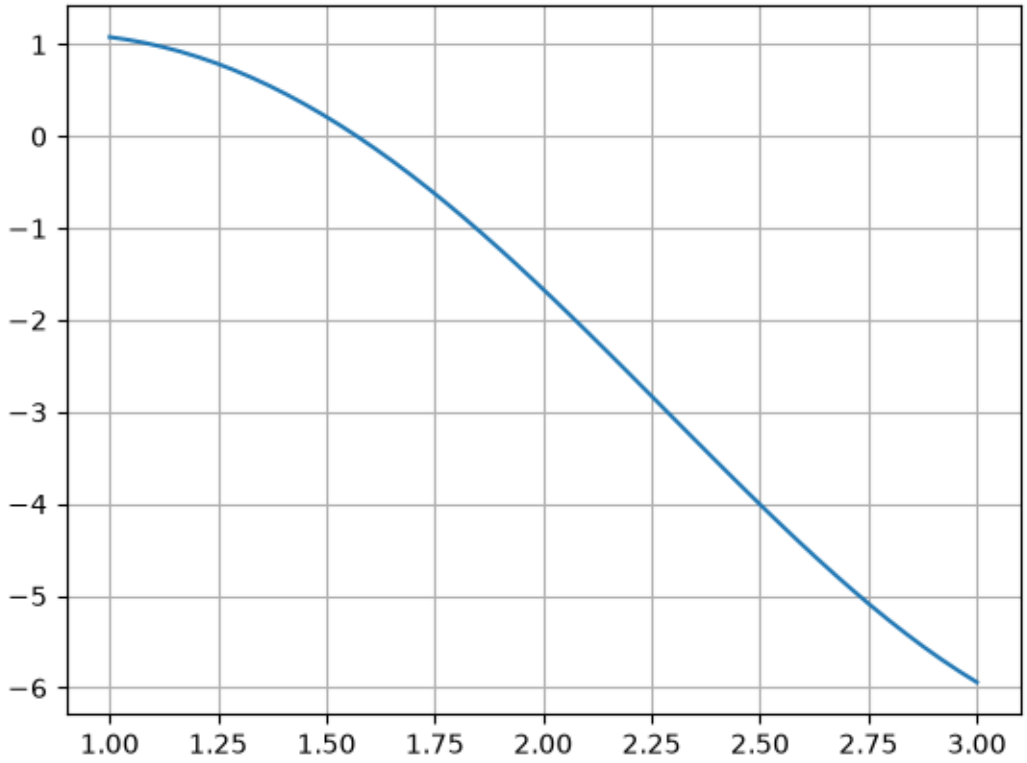
fx = sy.lambdify(x, fx_)

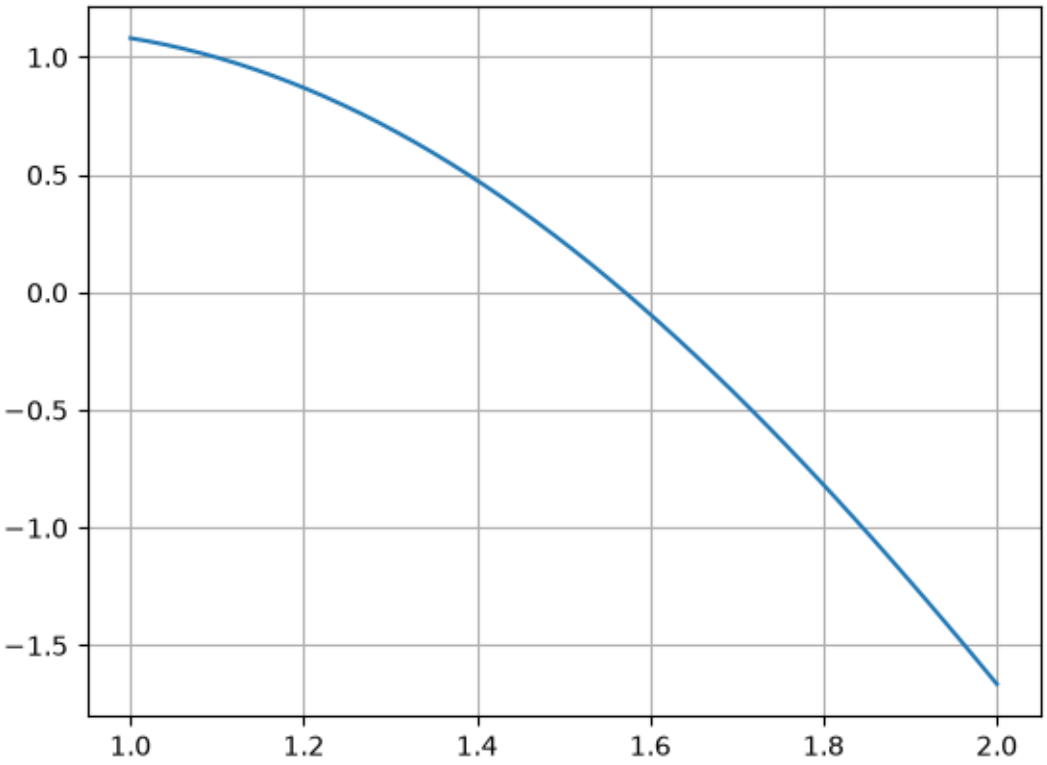
plot(fx, [-10, 10])
plot(fx, [0, 10])
plot(fx, [1, 3])
plot(fx, [1, 2])
```

$$f(x) = \frac{(2x^3 - 10x)\cos(x)}{x^2 - 5}$$

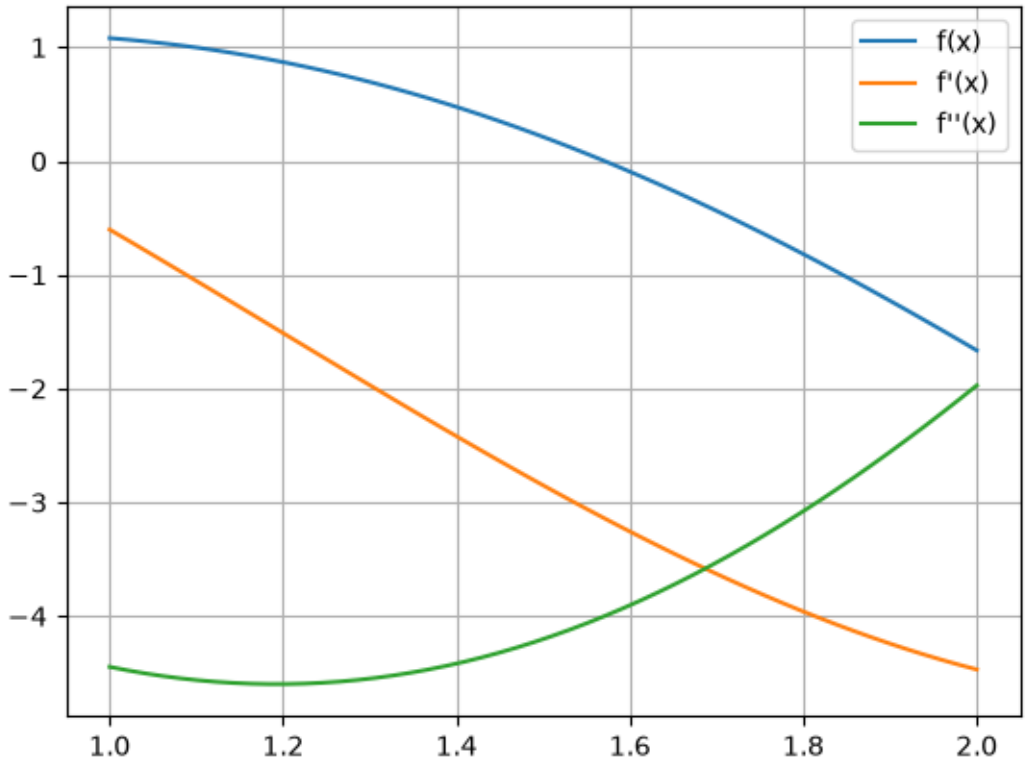



10
5
0
−5
−10
−15
−20
0
2
4
6
8
10


1
0
−1
−2
−3
−4
−5
−6
1.00
1.25
1.50
1.75
2.00
2.25
2.50
2.75
3.00



[287]:
```
intervalo = [1, 2]
plot3(fx_, intervalo)
```



[288]:
```
# Teste de convergência

# 1) f(x) é contínua com raiz no intervalo [a, b]
# 2) f'(x) != 0 para todo x em [a, b]
# 3) f''(x) não muda de sinal em [a, b]

# Passa pelo teste de convergência
raiz, iteracoes = newton_raphson(fx_, intervalo, 1e-5, 50)

display(iteracoes)

print('Raiz:', raiz)
# O método convergiu para a raiz
```

	k	x	f(x)	epsilon
0	1	1.500000	2.122116e-01	NaN
1	2	1.574434	-1.145396e-02	0.047277
2	3	1.570805	-2.629068e-05	0.002310
3	4	1.570796	-1.400628e-10	0.000005

```
Raiz: 1.57079632683948
```

1.1.5 e) $f(x) = x^6 - 7x^4 + 6x^2 - 1$

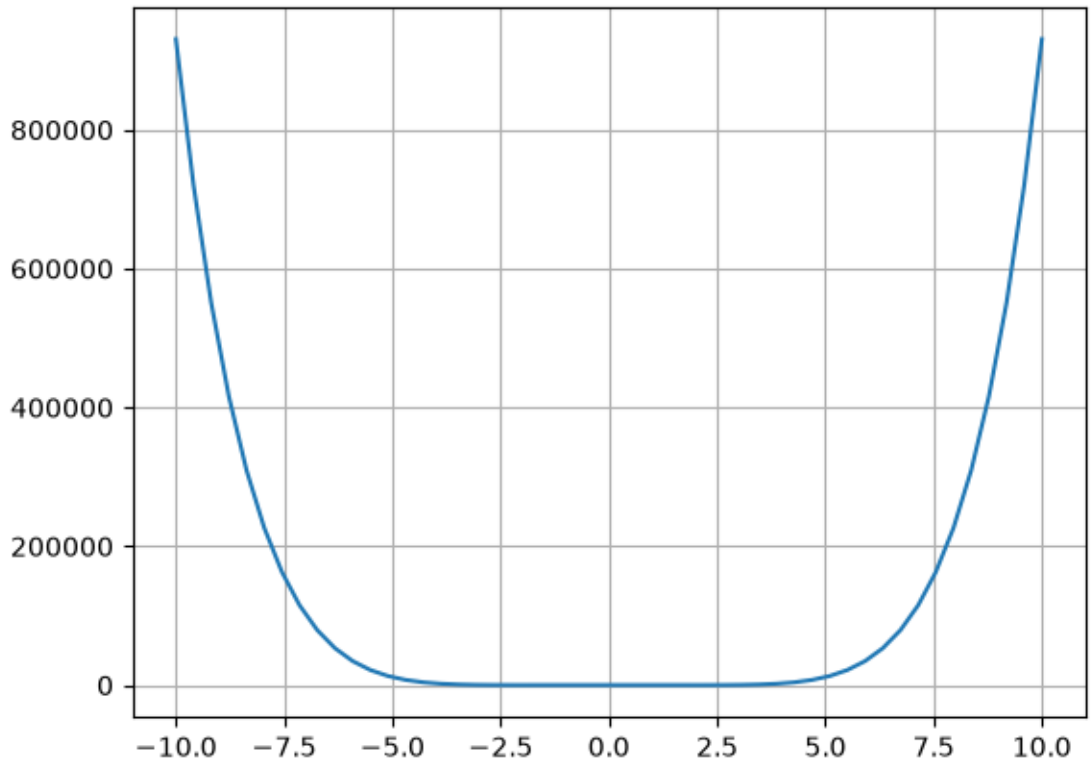
[289]:
```
# Definição da função
x = sy.Symbol('x')
fx_ = x**6 - 7*x**4 + 6*x**2 - 1

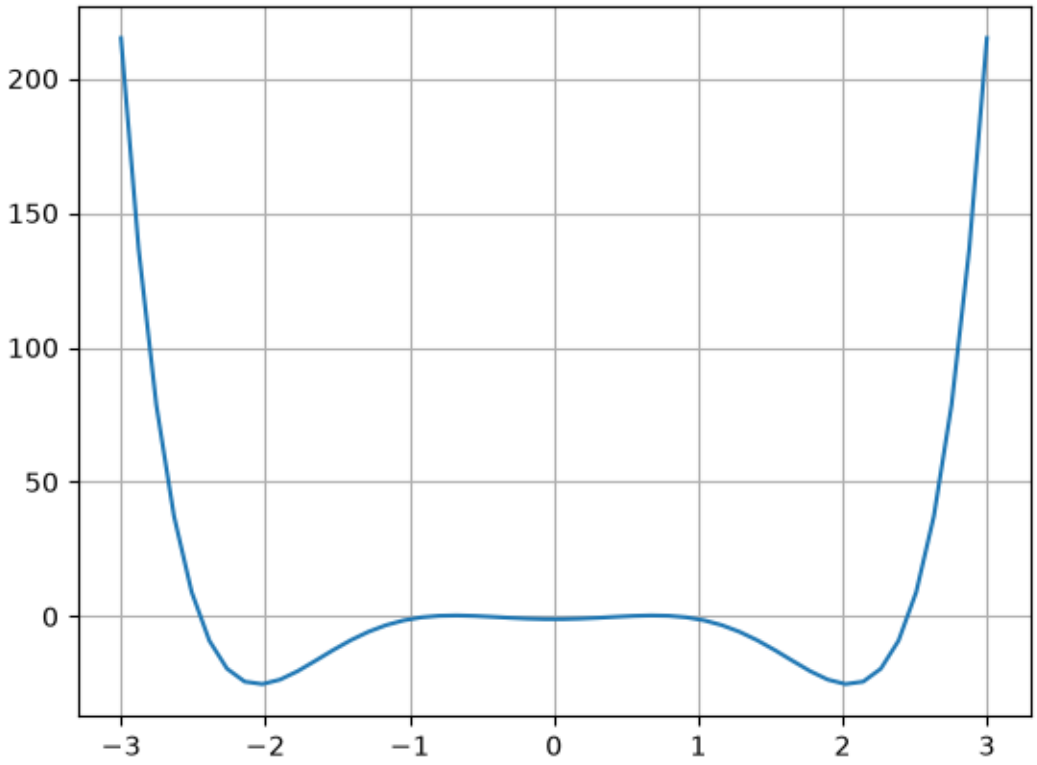
display(Math('f(x) = ' + sy.latex(fx_)))

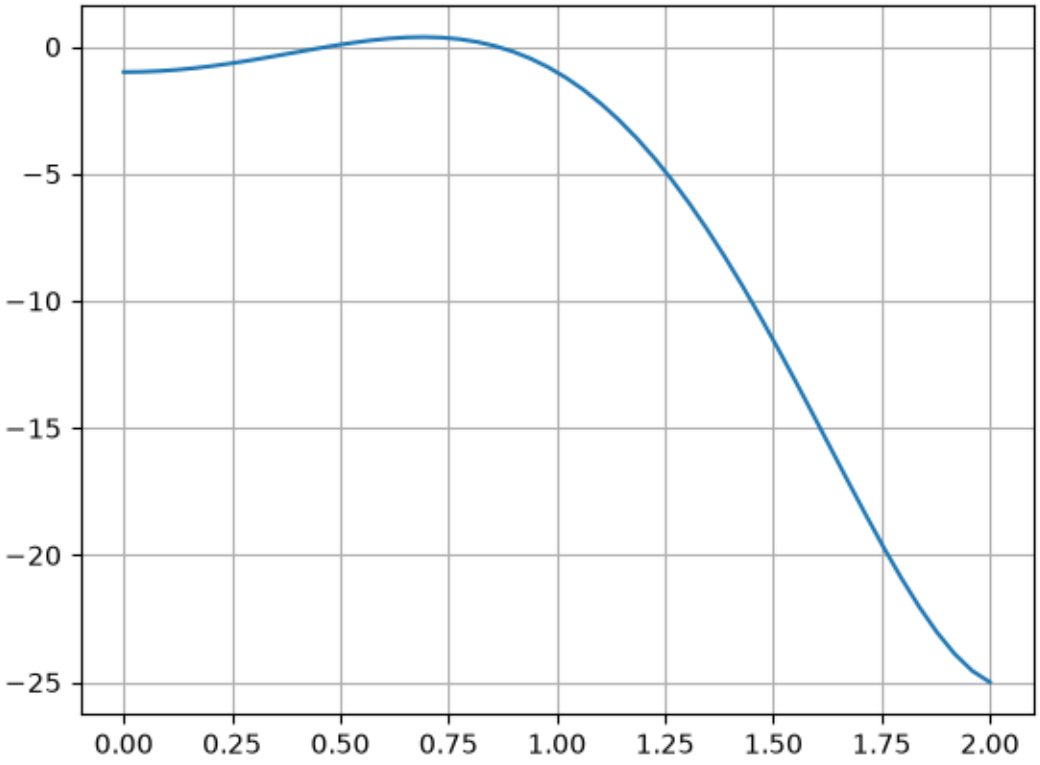
fx = sy.lambdify(x, fx_)

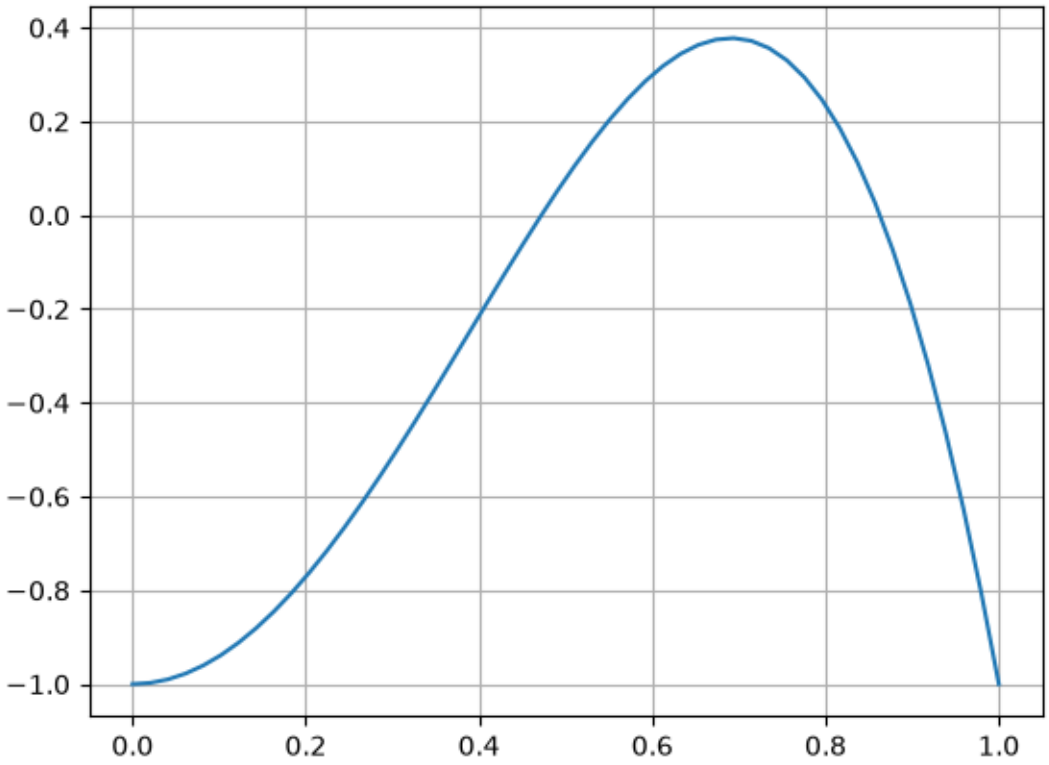
plot(fx, [-10, 10])
plot(fx, [-3, 3])
plot(fx, [0, 2])
plot(fx, [0, 1])
```

$f(x) = x^6 - 7x^4 + 6x^2 - 1$



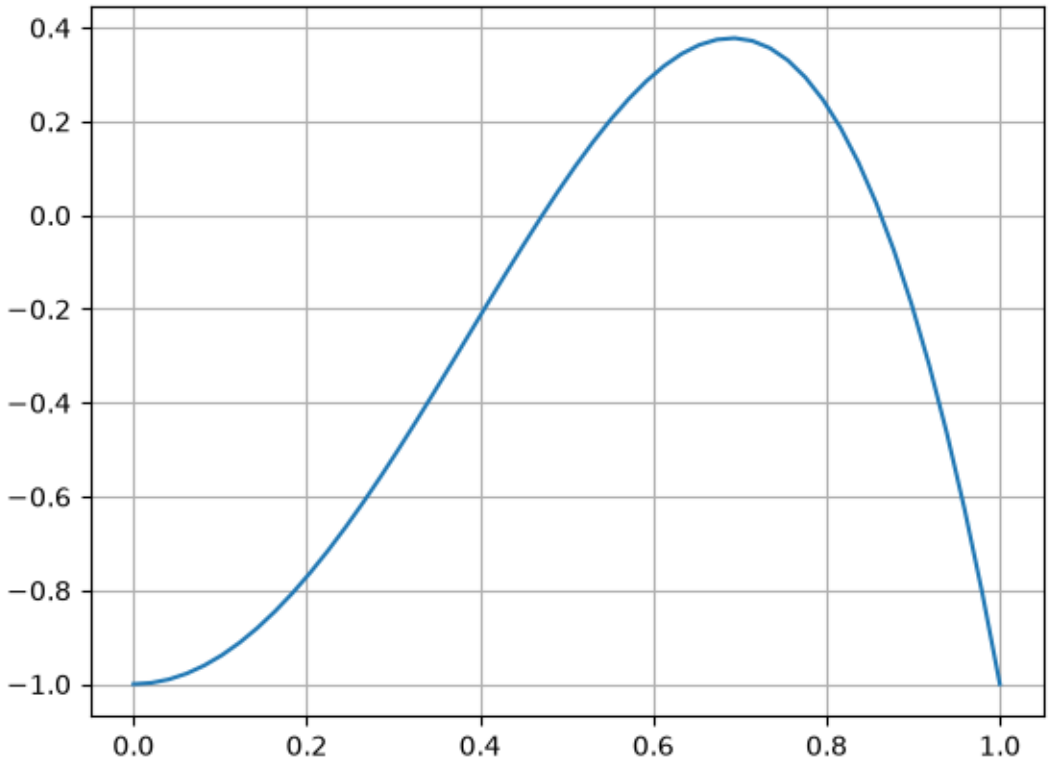

200
150
100
50
0
−3
−2
−1
0
1
2
3

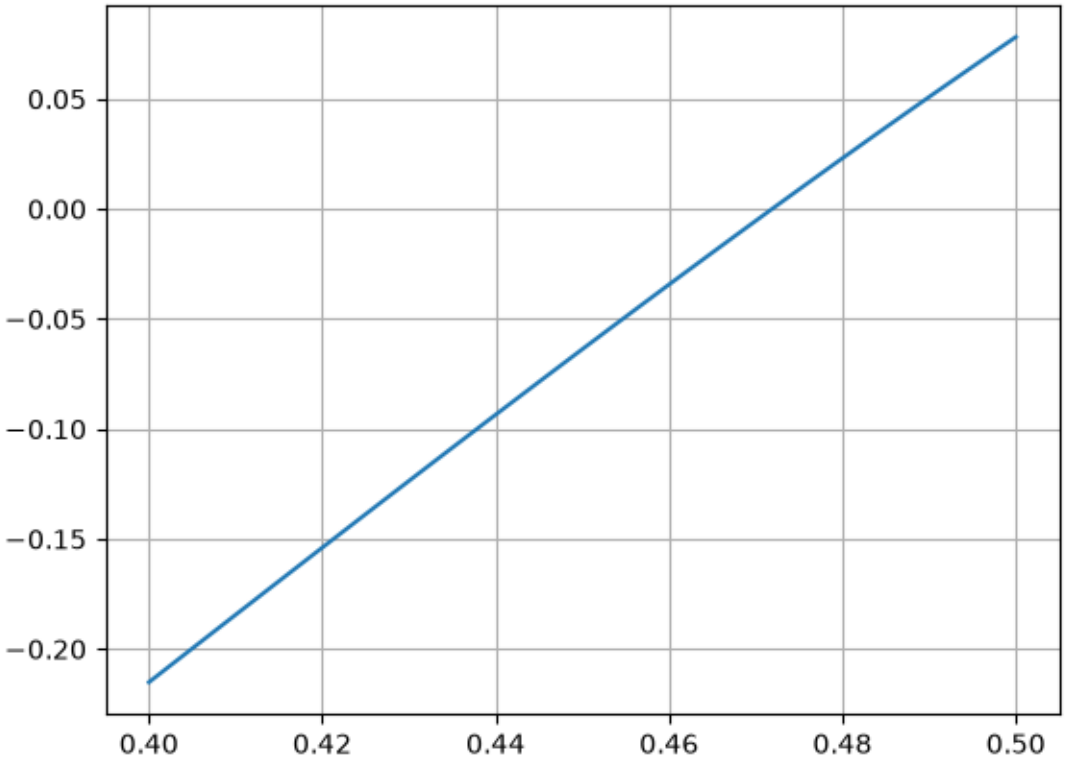

0
−5
−10
−15
−20
−25
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
2.00

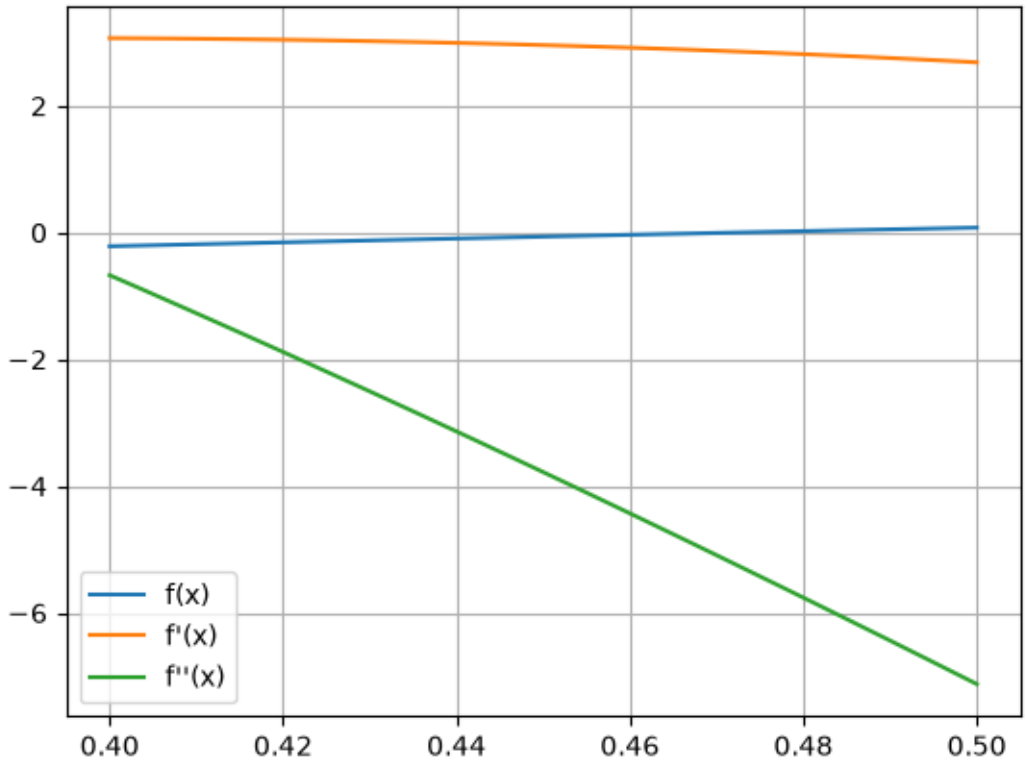


[290]:
```
plot(fx, [0, 1])
plot(fx, [0.4, 0.5])

intervalo = [0.4, 0.5]
plot3(fx_, intervalo)
```


0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
0.0
0.2
0.4
0.6
0.8
1.0


0.05
0.00
−0.05
−0.10
−0.15
−0.20
0.40
0.42
0.44
0.46
0.48
0.50



[291]:
```
# Teste de convergência

# 1) f(x) é contínua com raiz no intervalo [a, b]
# 2) f'(x) != 0 para todo x em [a, b]
# 3) f''(x) não muda de sinal em [a, b]

# Passa pelo teste de convergência
raiz, iteracoes = newton_raphson(fx_, intervalo, 1e-5, 50)

display(iteracoes)

print('Raiz:', raiz)
# O método convergiu para a raiz
```

	k	x	f(x)	epsilon
0	1	0.450000	-6.373998e-02	NaN
1	2	0.471539	-9.853625e-04	2.153948e-02
2	3	0.471884	-3.081645e-07	3.442139e-04
3	4	0.471884	-3.019807e-14	1.077177e-07

```
Raiz: 0.4718837988396693
```